

NGUYEN MINH KHOA

AI Engineer & Researcher

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PROFILE

Applied AI engineer and Research Assistant at AiTA Lab, FPT University, bringing model development, controlled experimentation and application engineering to AI product development. Experience spans text-to-molecule generation, mathematical expression recognition and LLM research workflows, connecting training data and evaluation with deployed inference services.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University

Oct 2024 - Present

- Authored and co-authored **five accepted/published papers**: HMER (*Mask CoMER*, *MTL + CBARM*), text-to-molecule generation (*LexiChem*, *ChemAligner-T5*) and VLM active learning (*CorrTie*).
- Selected published outcomes include up to **+5 ExpRate points** for Mask CoMER; ChemAligner-T5 reports **69.77% BLEU** and **31.28% Levenshtein distance** on L+M-24.

SELECTED PROJECTS

LexiChem | AI & Inference Contributor

2025 - 2026

Deployed molecular generation web application with FastAPI and NVIDIA Triton model serving.

- Built a three-dataset preprocessing pipeline covering **693,090 training examples** across L+M-24, Mol-Instructions, and ChEBI-20, with RDKit validation and canonicalization, deduplication, and SMILES-to-SELFIES conversion.
- Capstone ChEBI-20 experiments with shared-latent alignment improved BioT5+ from **87.2 to 88.8 BLEU**, **52.2 to 52.9 exact match**, and **90.7 to 93.5 MACCS FTS**, while maintaining **100% validity**.
- Integrated multi-model inference with FastAPI, NVIDIA Triton, RDKit, and MongoDB for molecule generation, validation, comparison, and saved experiment runs.

Research Ops

2026 - Present

Research infrastructure orchestrating literature understanding, experiment execution and evidence-backed manuscript development.

- Architected a FastAPI/PostgreSQL execution engine for configurable LLM research workflows, with persistent scientific state, execution budgets and human decision gates.
- Linked grounded LLM execution, experiment artifacts and manuscript evidence through versioned records and recovery.

NexOps | AI, Simulation & Platform Lead

2025 - Present

- Built a MES/SCADA prototype with repeatable machine/shift simulations and MQTT/EMQX telemetry through FastAPI and TimescaleDB for production planning, live monitoring, alarms and OEE.
- Integrated Bambu Lab printer control over LAN; coordinated hardware messaging with an IoT teammate.

SELECTED PUBLICATIONS (3 OF 5 ACCEPTED/PUBLISHED PAPERS)

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | Accepted and presented; publication pending

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pre-training and Regularization

ICDAR 2025 | ICORE 2026 Rank A | Published

[Paper](#) | [GitHub](#)

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | **SCIImago Q2 (2025)** | Published online Aug 2026

[Paper](#) | [GitHub](#)

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026

AWARDS & SCHOLARSHIPS

50% Tuition Scholarship, FPT University (2022) • **RESFES HCMC Second Prize** (2025, 2026) • **Demo Pitching Day** (VND 50M team award/project funding, 2025) • **Innovation Quest** (VND 40M team award/funding, 2025)

TECHNICAL SKILLS

AI & Research: Python, PyTorch, PyTorch Lightning, Hugging Face Transformers, scikit-learn, OpenCV, RDKit, ablation studies, error analysis, Weights & Biases, MLflow

Backend & Data: FastAPI, Pydantic, Pandas, PostgreSQL, TimescaleDB, MongoDB, Redis, MQTT/EMQX, REST APIs, NVIDIA Triton

Engineering & Reproducibility: Docker, Docker Compose, Linux, Git, Nginx, Playwright, LaTeX, scientific writing